Force Feedback Leader Arm for High Quality Demo Data using F/T Sensor

Chemin Ahn[1], Seunghwan Um[1], Sungjun Lee[2], Jeonghwan Park[1], Hyouk ryeol Choi[1,2*]
[1]Department of Mechanical Engineering, Sungkyunkwan University
[2]Department of Intelligent Robotics, Sungkyunkwan University

Portfolio

Introduction

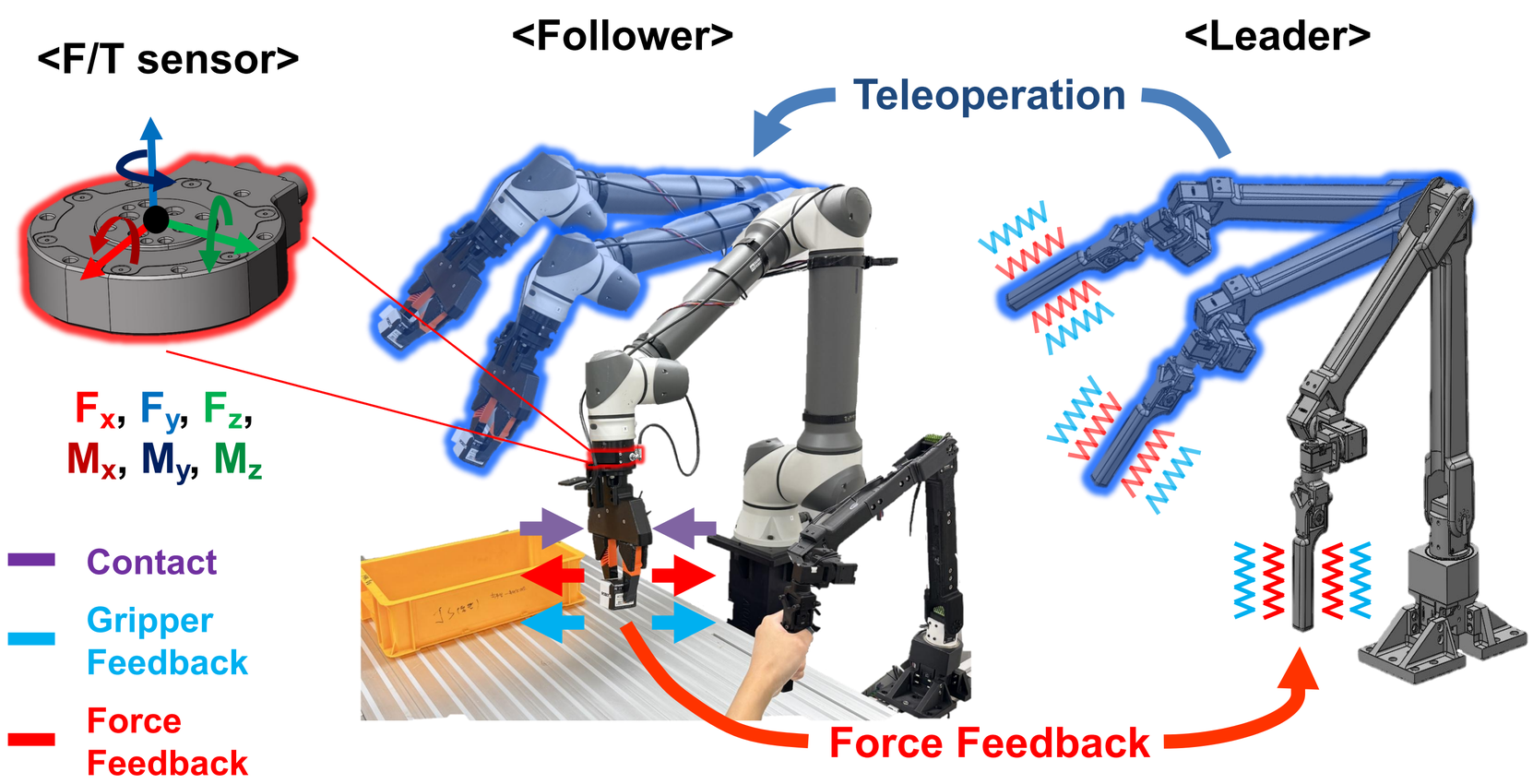
- **Why do we need high quality demo data?**
  - **Demo data quality = Imitation learning output (Action)**
  - **Vision sensors have inherent limitations**
    Lack of physical interaction data

Explain & Results

- **Force Feedback via F/T sensor**
  - $\tau_{Feedback} = k_f(J^T F_{ext})$:
    Senses **contact through leader arm's force feedback**
- **Gravity Compensation**
  - $\tau_{Gravity} = k_g(J^T F_{gravity})$:
    Compensates for the leader arm's self-weight
- **Teleoperation**
  - Leader current joint angles = Follower target joint angles
  - Teleoperation w/ **feedback and gravity compensation**

Results

1. Faster and easier collection of **high-quality demo data**
2. More **intuitive** and immersive manipulation
3. Utilization of **force data for training**
4. Precise control
5. System **scalability** & cost reduction



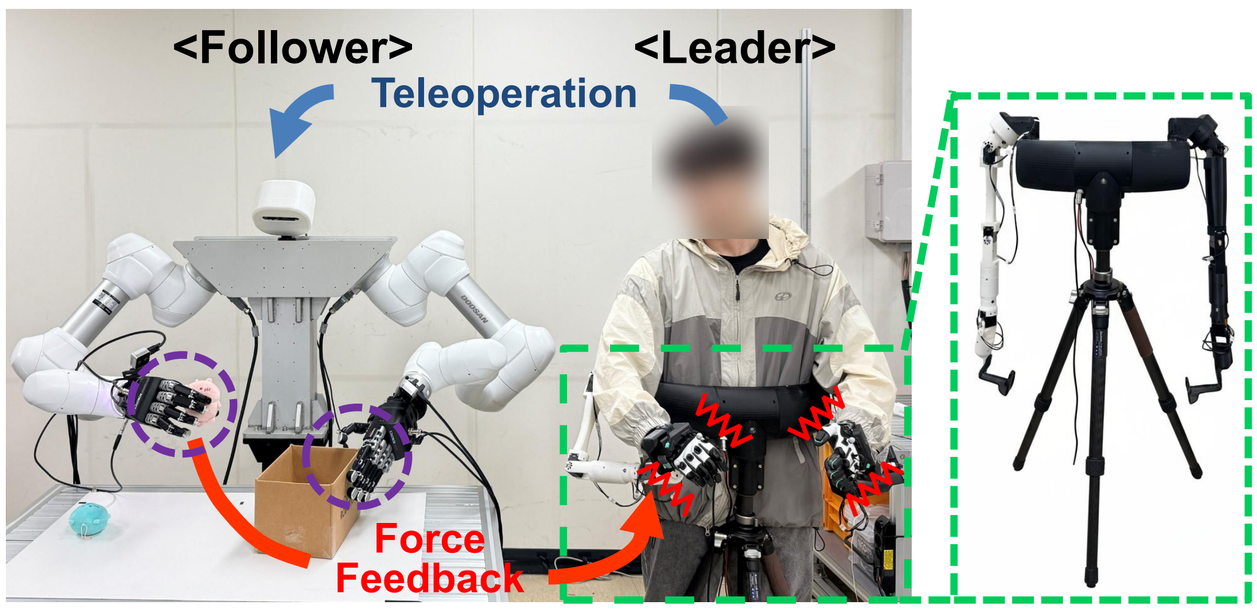
Leader - Follower System & Force Feedback

Functions

1. **Force Feedback**
   enabled on all joints
   (Follower → Leader)
2. **Gravity compensation**
   enabled on all joints
3. **Teleoperation**
   (Leader → Follower)

Applications

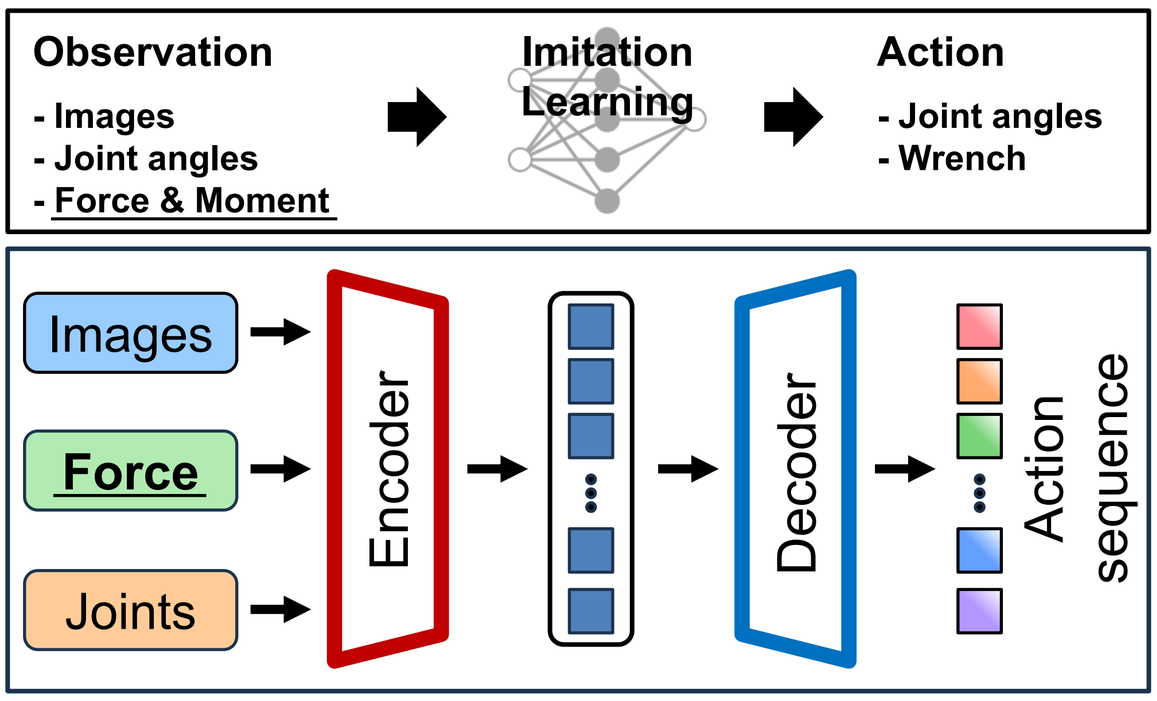
- **Modulization**
  - Force feedback and data acquisition can be implemented **without altering the existing setup**, simply by adding an F/T sensor.
  - **Easily scalable & Easy to set** the system.



Dual arm Leader - Follower system

Dual arm Leader arm

- **Multimodal Imitation Learning**
  - Force data can be used to train the policy
  - Learn policy from high-quality demos
  - Learn policy with contact-awareness



This project was funded by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government(MOTIE)(P0026249)

Robotics Innovatory · 성균관대학교 SUNGKYUNKWAN UNIVERSITY · 산업통상부 Ministry of Trade, Industry and Resources · beyond leading technology KIAT Korea Institute for Advancement of Technology